

Higher Nationals - Summative Assignment Feedback Form

Student Name/ID	Subakar Habinan		
Unit Title	Unit 02: Networking		
Assignment Number	1	Assessor	
Submission Date		Date Received 1st submission	
Re-submission Date		Date Received 2nd submission	

Assessor Feedback		
Grade:	Assessor Signature:	Date:
Resubmission Feedback: Please note resubmission feedback is focussed only on the resubmitted work		
Grade:	Assessor Signature:	Date:
Internal Verifier's Comments:		
Signature & Date:		

- Please note that grade decisions are provisional. They are only confirmed once internal and external moderation has taken place and grades decisions have been agreed at the assessment board.

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4. Ensure that you give yourself enough time to complete the assignment by the due date.
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6. You must take responsibility for managing your own time effectively.
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13. Use **footer function in the word processor to insert Your Name, Subject, Assignment No, and Page Number on each page**. This is useful if individual sheets become detached for any reason.

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When submitting evidence for assessment, each student must sign a declaration confirming that the work is their own.

Student name:		Assessor name:	
Issue date:	Submission date:	Submitted on:	
Programme: Pearson BTEC HND in computing			
Unit: Unit-02 Networking			
LAN Design & Implementation for Redco Development.			

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Student Declaration

Student declaration	
I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.	
Student signature:	Date:

Unit 2 – Networking Assignment Brief

Student Name/ID Number	
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Unit Number and Title	Unit 2 – Networking
Academic Year	2024/2025
Unit Tutor	
Assignment Title	LAN Design & Implementation for Redco Developments.
Issue Date	
Submission Date	
Submission Format	
The assignment submission is in the form of the following. Presentation - A formal 10–15-minutes presentation (8-10 slides as a guide, with supporting speaker notes) to communicate an investigation to a non-technical audience discussing the key features and characteristics of a range of network types, topologies, hardware, and software that you have been used for the new network implementation. This should demonstrate your network blueprint and the packet tracer simulation. A Written report- The submission should be in the form of an individual report written in a concise, formal business style using single spacing and font size 12. You are required to make use of headings, paragraphs, and subsections as appropriate, and all work must be supported with research and referenced using Harvard referencing system. Please also provide an end list of references using the Harvard referencing system. The recommended word count is 3,000–3,500 words for the report excluding annexures, although you will not be penalised for exceeding the total word limit. Note: Please Don't add any ZIP files to the ELMS it should be a PDF document with relevant could links to the packet tracer files.	
Unit Learning Outcomes	
LO1. Examine networking principles and their protocols. LO2. Explain networking devices and operations. LO3. Design efficient networked systems. LO4. Implement diagnose and demonstrate prepared networked systems.	
Transferable skills and competencies developed	

- Computational thinking (including its relevance to everyday life)
- Demonstrate knowledge and understanding of essential facts, concepts, principles, and theories relating to computing and computer applications.
- Use such knowledge and understanding in the modelling and design of computer-based systems for the purposes of comprehension, communication, prediction, and the understanding of trade-offs.
- Recognize and analyze criteria and specifications appropriate to specific problems, and plan strategies for their solutions.
- Critical evaluation and testing: analyze the extent to which a computer-based system meets the criteria defined for its current use and future development.
- Methods and tools: deploy appropriate theory, practices and tools for the design, implementation, and evaluation of computer-based systems.

Computing-related practical skills:

- The ability to specify, design and construct reliable, secure, and usable computer-based systems.
- The ability to evaluate systems in terms of quality attributes and possible trade-offs presented within the given problem.
- The ability to deploy effectively the tools used for the construction and documentation of computer applications, with particular emphasis on understanding the whole process involved in the effective deployment of computers to solve practical problems.
- The ability to critically evaluate and analyze complex problems, including those with incomplete information, and devise appropriate solutions, within the constraints of a budget.

Generic skills for employability

- Intellectual skills: critical thinking; making a case; numeracy and literacy.
- Self-management: self-awareness and reflection; goal setting and action planning.
- Independence and adaptability; acting on initiative; innovation and creativity.
- Interaction: reflection and communication.
- Contextual awareness, e.g. the ability to understand and meet the needs of individuals, business, and the community, and to understand how workplaces and organisations are governed.

OBSERVATION RECORD

Learner name:	
Qualification:	
Unit number & title:	
Description of activity undertaken	

Assessment criteria

How the activity meets the requirements of the assessment criteria

Learner name:			
Learner signature:		Date:	
Assessor name:			
Assessor signature:		Date:	

WITNESS STATEMENT

Learner name:			
Qualification:			
Unit number & title:			
Description of activity undertaken (please be as specific as possible)			
Assessment criteria (for which the activity provides evidence)			
How the activity meets the requirements of the assessment criteria, including how and where the activity took place			
Witness name:		Job role:	
Witness signature:		Date:	
Learner name:			
Learner signature:			Date:
Assessor name:			
Assessor signature:			Date:

Assignment Brief and Guidance:	

Redco Developments is a Sri Lankan based Mobile app development company, which the head office is in Kandy. The company develops mobile applications for Sri Lanka's leading organizations for both mobile platforms IOS and Android.

The CEO plans to expand the company operations of Redco development in Southern province in Sri Lanka. The management has decided to open a new branch office in Galle and wants it to be one of the most prominent tech-oriented offices in Galle with the latest tech facilities including smart devices such as auto lighting, physical security solutions, smart gates and new ERP software IP camera Systems.

You have been appointed as the Junior implementation Engineer of Redco and your task is to design a new network for the Galle branch and restructure the existing Kandy network that connects both networks.

Prepare a network architectural design and implement it with your suggestions and recommendations to meet the company requirements.

The floor plan of the head office in Kandy is as follows:

Floor 1:

- Reception area (2 employees)
- Sales & Marketing Department (10 employees)
- Customer Services Area – with Wi-Fi facilities
- Developers (25 employees)

Floor 2:

- Director suits (3 suits)
- Boardroom with Video conferencing facility and Wi-Fi.
- Administration Department (15 Employees)
- HR Department (4 employees)

Floor 3:

- Accounting & Finance Department (12 employees)
- IT Technical support (4 employees)
- The Server Room

The newly established floor plan of the Galle is as follows:

Floor 1:

- Reception area (2 employees)
- Customer Services Area– with Wi-Fi facilities
- Developers (40 employees)
- 5 IP cameras

Floor 2:

- Administration Department (10 Employees)
- HR Department (5 employees)
- Accounting & Finance Department (12 employees)
- IT Department (5 employees)
- The Server Room
- 6 IP cameras

The following requirements are given by the Management.

- All the departments **must be separated** with **unique subnets**.
- **The conference room of the head office and Customer Services Areas** of each branch are to be **equipped with Wi-Fi connections with the security enabled**.
- **Connectivity between two branches** (Kandy & Galle) which would allow intra-branch connectivity between departments. (Use Inter VLAN routing RIP or any routing protocol- evidence must be provided)
- **IP range for 200.100.10.0/24** for Kandy and **200.100.20.0/24** for Galle branch and subnetting calculations must be provided within the report **except the server room**.
- **All the PC must retrieve IP from the DHCP server**.
- **The number of servers required for the Server room** needs to be decided by the Network designer and should be assigned with **200.100.50.0/24** subnet. (Uses **static IPs**)
- **The Sales and Marketing Team** also needs to access Network resources **using**

WIFI connectivity.

(Note: Clearly state your assumptions. You are allowed to design the network according to your assumptions, but the main requirements should not be violated)

Activity 01

- Discuss the benefits and constraints of different network system types that can be implemented in the Galle branch as well as for the restructuring process what type of network recommended for the Galle and Kandy offices.
- Discuss the different protocols utilized for communication and the connectivity of the 2 offices which can be implemented in the Galle branch and the How the OSI and TCP model influencing in designing network for Galle office.
- Discuss the importance and impact of network topologies that are used in network design using examples.
- Recommend suitable network topologies and network protocols for the above scenario and evaluate your answer with valid points.

Activity 02

- Discuss the operating principles of network devices (Ex: Router, Switch, Etc.).
- Explain different server types that can be used for the above scenario while exploring different servers that are available in today's market with their specifications. You should recommend a server/server for the above scenario and justify your selection with valid points.
- Discuss the inter-dependence of workstation hardware and networking software, networking operating systems you use based on the scenario. Provide examples for networking software that can be used in the above network design.

Activity 03

- Prepare a written network design plan to meet the above-mentioned user requirements including a blueprint drawn using a modeling tool (Ex: Microsoft Visio, EdrawMax). Test and evaluate the proposed design by analysing user feedback with the aim of optimizing your design and improving efficiency.

- Support your answer by providing the VLAN and IP subnetting scheme for the above scenario and the list of devices, network components and software used to design the network for the above scenario and while justifying your selections.
- Install and configure Network services, devices and applications
Ex: VLAN, WiFi, DNS, Proxy, Web, Etc
According to the proposed design to accomplish the user requirements. Design a detailed Maintenance schedule for the above Network.

***Note: - Screen shots of Configuration scripts should be presented in your document. Your packet tracer file must be submitted to the cloud drive and share the link in the appendix of document for the assessor verification purposes.**

Don't add any ZIP files to the ELMS it should be a PDF document with relevant could links to the packet tracer files

Activity 04

- Implement a networked system based on your prepared design with valid evidence.
- Develop test cases and conduct verification (Ex: Ping, extended ping, trace route, telnet, SSH, etc.) to test the above Network and analyse the test results against the expected results.
- Recommend potential future enhancements for the networked system with valid justifications and critically reflect on the implemented network, including the plan, design, configurations, tests, and the decisions made to enhance the system.

Activity 05 – Presentation

- Prepare and present 10-15 minutes presentation which includes the design (Network type, topologies, devices, and software you used for the implementation. You need to justify your design with the valid reasons.

- The presentation should include,
 - Blueprint of the design
 - Packet tracer demonstration
 - Testing interface pinging's
 - Trace routes

Note: Add your presentation slides in the appendix of the report.

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Examine networking principles and their protocols		LO1 and LO2 D1 Evaluate the topology and protocol suite selected for a given scenario and how it demonstrates the efficient utilisation of a networking system.
P1 Discuss the benefits and constraints of different network types and standards. P2 Explain the impact network topologies have on communication and bandwidth requirements.	M1 Assess common networking principles and how protocols enable the effectiveness of networked systems.	
LO2 Explain networking devices and operations		
P3 Discuss the operating principles of networking devices and server types. P4 Discuss the interdependence of workstation hardware and relevant networking software.	M2 Explore a range of server types and justify the selection of a server for a given scenario, regarding cost and performance optimisation.	
LO3 Design efficient networked systems		LO3 and LO4 D2 Critically reflect on the implemented network, including the design and decisions made to enhance the system.
P5 Design a networked system to meet a given specification. P6 Design a maintenance schedule to support the networked system.	M3 Analyse user feedback on your designs with the aim of optimising your design and improving efficiency.	
LO4 Implement and diagnose networked systems		
P7 Implement a networked system based on a prepared design. P8 Document and analyse test results against expected results.	M4 Recommend potential enhancements for the networked systems.	

Recommended resources

Please note that the resources listed are examples for you to use as a starting point in your research – the list is not definitive.

Weblinks:

- <https://blog.netwrix.com/> (2022) Network Devices Explained [online] Available at: <https://blog.netwrix.com/2019/01/08/network-devices-explained/> [Accessed 1 August 2022]
<https://www.checkpoint.com/> (n.d.) What is a Firewall? [online] Available at: <https://www.checkpoint.com/cyber-hub/network-security/what-is-firewall/> [Accessed 1 August 2022]
- <https://www.checkpoint.com/> (n.d.) What is an IoT Gateway? [online] Available at: <https://www.checkpoint.com/cyber-hub/network-security/what-is-iot/what-is-an-iot-gateway/> [Accessed 1 August 2022]
- <https://www.cloudflare.com/en-gb/> (n.d.) A global network built for the cloud [online] Available at: <https://www.cloudflare.com/en-gb/learning/network-layer/internet-protocol/> [Accessed 1 August 2022]
- <https://www.comparitech.com/> (2020) Variable Length Subnet Mask (VLSM) Tutorial [online] Available at: <https://www.comparitech.com/net-admin/variable-length-subnet-mask-vlsm-tutorial/> [Accessed 1 August 2022]
- <https://www.comptia.org/> (n.d.) What Is a Network Protocol, and How Does It Work? [online] Available at: <https://www.comptia.org/content/guides/what-is-a-network-protocol> [Accessed 1 August 2022]
- <https://www.ibm.com/uk-en> (2021) Networking [online] Available at: <https://www.ibm.com/uk-en/cloud/learn/networking-a-complete-guide> [Accessed 1 August 2022]
- <https://www.ibm.com/uk-en> (2022) TCP/IP protocols [online] Available at: <https://www.ibm.com/docs/en/aix/7.2?topic=protocol-tcpip-protocols> [Accessed 1 August 2022]
- <https://www.lifewire.com/> (2022) What Is Bandwidth? Definition, Meaning, and Details [online] Available at: <https://www.lifewire.com/what-is-bandwidth-2625809> [Accessed 1 August 2022]
- <https://www.ncsc.gov.uk/> (2019) Secure design principles [online] Available at: <https://www.ncsc.gov.uk/collection/cyber-security-design-principles> [Accessed 1 August 2022]
- <https://www.serverwatch.com/> (2021) Network Server [online] Available at: <https://www.serverwatch.com/servers/network-server/> [Accessed 1 August 2022]
- <https://www.techtarget.com/> (2022) IoT gateway [online] Available at: <https://www.techtarget.com/iotagenda/definition/IoT-gateway> [Accessed 1 August 2022]
- <https://www.univention.com/> (2022) Brief Introduction: DHCP and DNS [online] Available at: <https://www.univention.com/blog-en/brief-introduction/2019/03/brief-introduction-dhcp-dns/> [Accessed 1 August 2022]

Virtual Network Simulators:

- <https://www.adobe.com/> (n.d.) DNS/DHCP/EMAIL VIA PACKET TRACER [online] Available at: <https://express.adobe.com/page/7ogipygZfOh0B/> [Accessed 1 August 2022]
- <https://techgenix.com/> (2019) Tips and tools for simulating a complex network in a virtual lab [online] Available at: <https://techgenix.com/simulating-network-in-virtual-lab/> [Accessed 1 August 2022]

- <https://www.eve-ng.net/> (2022) EVE - The Emulated Virtual Environment For Network, Security and DevOps Professionals [online] Available at: <https://www.eve-ng.net/> [Accessed 1 August 2022]
- <https://www.gns3.com/> (2022) The software that empowers network professionals [online] Available at: <https://www.gns3.com/> [Accessed 1 August 2022] <https://www.netacad.com/> (n.d.) Cisco Packet Tracer [online] Available at: <https://www.netacad.com/courses/packet-tracer> [Accessed 1 August 2022]

Journal articles.

- Agyemang, J., Kponyo, J. and Klogo, G., 2022. The State of Wireless Routers as Gateways for Internet of Things (IoT) Devices. [online] Pubs.sciepub.com. Available at: [Accessed 1 August 2022].
- Oje, A. (2021) Optimization and analysis of the packet switched network with focus on the 3G network. Journal of Physics: Conference Series, Volume 1734, International Conference on Recent Trends in Applied Research doi:10.1088/1742-6596/1734/1/012037 Available at: <https://iopscience.iop.org/article/10.1088/1742-6596/1734/1/012037/meta> [Accessed 1 August 2022].
- Tyagi, A. (2020) TCP/IP Protocol Suite. International Journal of Scientific Research in Computer Science Engineering and Information Technology doi:10.32628/CSEIT206420 Available at: https://www.researchgate.net/publication/346829282_TCPIP_Protocol_Suite [Accessed 1 August 2022].
- Van der Toorn et al. (2022) Addressing the challenges of modern DNS a comprehensive tutorial. Computer Science Review, Volume 45, 2022, 100469, <https://doi.org/10.1016/j.cosrev.2022.100469>
- Xu, G. (2021) Research on the Application of the IPv6 Network Protocol. Journal of Physics: Conference Series doi:10.1088/1742-6596/2031/1/012040 Available at: <https://iopscience.iop.org/article/10.1088/1742-6596/2031/1/012040/pdf> [Accessed 1 August 2022].
- Bonaventure, O. (2011) Computer Networking: Principles, Protocols and Practice, The Saylor Foundation, Available at: <https://resources.saylor.org/wwwresources/archived/site/wp-content/uploads/2012/02/Computer-Networking-Principles-Bonaventure-1-30-31-OTC1.pdf>

Grading Criteria	Achieved	Feedback
LO1: Examine networking principles and their protocols		
P1 Discuss the benefits and constraints of different network types and standards.		
P2 Explain the impact of network topology, communication, and bandwidth requirements.		
M1 Assess common networking principles and how protocols enable the effectiveness of networked systems.		
LO2: Explain networking devices and operations.		
P3 Discuss the operating principles of networking devices and server types.		

P4 Discuss the interdependence of workstation and software relevant networking software		
M2 Explore a range of server types of hardware configuration and justify the selection of a server for a given scenario, regarding cost and performance optimization.		
D1 Evaluate the topology, protocols and networking software selected for a given scenario and how it demonstrates the efficient utilization of a networking system.		
LO3: Design efficient networked systems		
P5 Design a networked system to meet a given specification.		

P6 Design a maintenance schedule to support the networked system.		
M3 Analyze user feedback on your designs with the aim of optimizing your design and improving efficiency of the new network implementation.		
D2 Critically reflect on the implemented network, including the design and decisions made to enhance the system.		
LO4: Implement and diagnose networked systems.		
P7 Implement a networked system based on a prepared design.		
P8 Present the network types of topologies and the blueprint to the audience with simulation testing.		

M4 Recommend potential enhancements for the networked systems.		
D2 The PowerPoint and the packet tracer demonstration fulfil the pinging, extended pinging trace root and telnet or SSH in the implementation. Document test results against expected results (ping tests and DCHP results).		

Acknowledgement

I would like to express my deepest gratitude to everyone who contributed to the successful completion of this project.

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Thank you.

Subakar Habinan
HND in Network Engineering
E-soft Metro Campus

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Activity 01

Introduction to Computer Networking

Computer networking is the act of linking a single or more computers plus computer equipment to facilitate communication and exchange of information or resources among the computers or computing equipment. It entails application of hardware, software, protocols, and standards to design and sustain networks through which they are able to share files, printers, internet, and ERP systems.

Benefits and Constrains of Computer networks

There are many benefits and constrains in computer networks,

Benefits	Constrains
Resource sharing - Networks let devices share things like printers, scanners, and storage. This saves time and money.	Reliability - Networks depend on hardware and software. If something fails, the network may stop working.
Communication - Devices can easily communicate with each other. People can share information and work together faster.	Interoperability - Different devices may not work well together. This can cause problems in communication.
Remote access - Users can access files and resources from anywhere. This helps people work from home or while traveling.	Cost - Setting up and maintaining a network can be expensive, especially for large systems.
Increased productivity - Devices work together efficiently. This helps people complete tasks faster.	Complexity - Setting up and maintaining a network can be expensive, especially for large systems.
Backup and disaster recovery - Networks help store backups and recover data if something goes wrong. This keeps important data safe	Security - Networks can be attacked by hackers. This can lead to data loss or unauthorized access.

Table 1: computer network benefits and constraints

Network Architecture model

A network architecture is a blueprint that defines the structure, components, and communication protocols of a network. It serves as a guide for designing, implementing, and maintaining a network.

Peer to Peer Architecture

Peer-to-peer (P2P) is a type of network where all devices are equal

- Each device (called a peer) works as both a client and a server
- Devices can share resources directly with each other
- There is no central server controlling the network

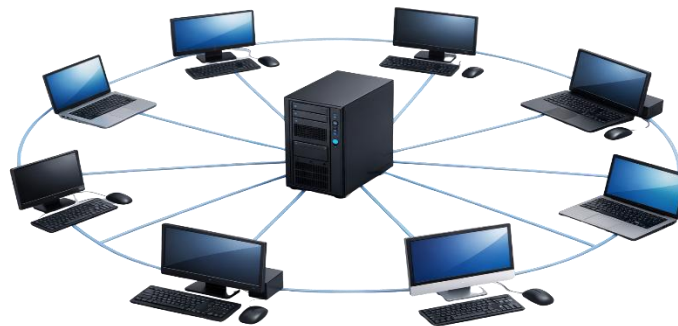


Figure 1: Peer-to-peer network

Client Server Architecture

A client-server architecture is a type of network where devices are divided into two groups: clients and servers. The server is a powerful computer that provides resources and services, while the clients are devices that request and use those services. Clients send requests to the server, and the server responds by providing the needed data or service. This model is widely used because it allows different parts of a system to work together, even if they are in different locations.

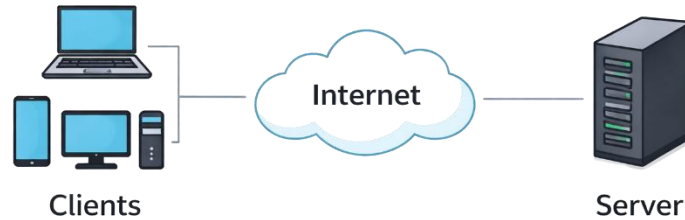


Figure 2: client-server architecture network

Comparison between peer to peer and client-server architectures

Peer-to-Peer Architecture	Client-server architectures
P2P networks are more robust and resilient, since the failure of one device will not affect the entire network	Clients can access server resources from anywhere, remotely
Each device can act as both a client and a server.	Server is responsible for data consistency, integrity, and security
Number of devices can be increased easily	Services are provided by the server and consumed by clients
Devices can share and access resources directly with each other.	Better control and centralized management of resources and services
The network doesn't rely on a single device, so there's no single point of failure.	New servers can be added to support more clients and handle extra load.
Adding more devices is easy.	The server provides services that clients use.
No single point of failure	New servers can be added to handle more clients and extra load

Table 2: Comparison between peer to peer and client-server architectures

Cloud Computing

Cloud computing is a way of providing computing resources like servers, storage, and software over the internet. Users can access and use these resources whenever they need them, without having to set up or manage their own hardware or infrastructure.

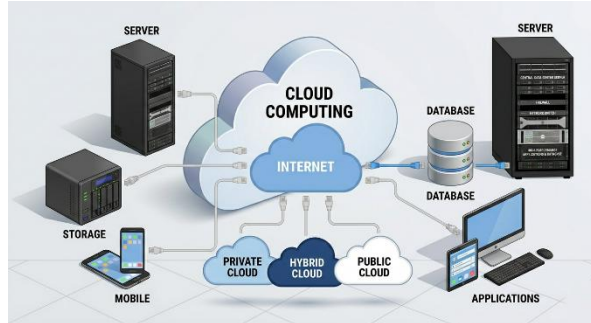


Figure 3: Cloud Computing

There are three main types of cloud computing,

1. Infrastructure as a Service (IaaS): Lets you use virtual computers, storage, and other resources online without owning the physical hardware.
2. Platform as a Service (PaaS): Lets you build, run, and manage applications online without worrying about the underlying servers or systems.
3. Software as a Service (SaaS): Lets you use software and apps over the internet without installing them on your own device.

Many companies provide cloud services, including Amazon, Microsoft, Google, IBM, Oracle, and others.

Virtualized Systems

Virtualized networks, also called SDNs, separate the control and data parts of the network. Normally, in traditional networks, both control (deciding where data goes) and data (actually sending the data) are handled by the same devices, like routers and switches. In virtualized networks, the control part runs in software on servers, while the data part stays in the hardware devices.

Centralized Systems

A centralized network system is a type of network where one main controller manages everything. This controller is usually a server or a group of servers kept in one place, like a data center. It controls the network devices, such as routers and switches, and makes sure all the network rules are followed.

Types of networks

There are many types of networks currently in use. Some of them are,

- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)
- Campus Area Network (CAN)
- Storage Area Network (SAN)
- Virtual Private Network (VPN)

Local Area Network (LAN)

A LAN is a network that connects computers and devices in a small area, like a home, office, or building. It can use wired technologies like Ethernet, Token Ring, and FDDI, or wireless technologies like Wi-Fi. LANs allow users to share resources such as printers, scanners, and files, and make it easy for people to communicate and work together.

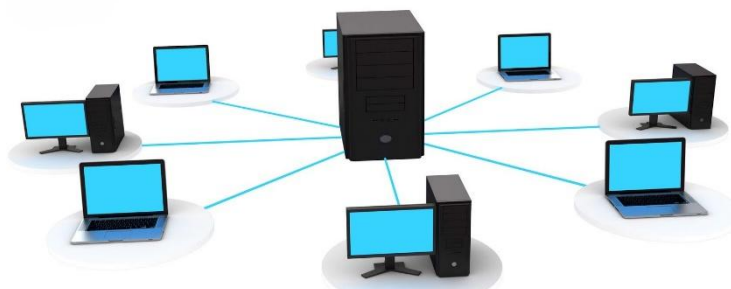


Figure 4: Local Area Network (LAN)

Benefits and constrains of LAN

Benefits	Constrains
LAN allows devices to share resources like printers, scanners, and files. This helps save money and improves productivity.	LAN usually needs devices to be physically connected, so it reduces mobility and flexibility.
Users can communicate and work together in real-time, which makes work faster and more efficient.	It only covers a small area like a building or office, so it cannot connect devices over long distances.
LAN is easy to manage because all devices are connected in one small area. Network administrators can easily monitor and fix problems.	LAN can face security risks like hacking, malware, and phishing if it is not properly managed.
LANs are low cost to set up and maintain because they use common hardware and software.	LANs may have limited bandwidth, which can make the network slow at times.
LANs can be made secure using tools like firewalls, encryption, and intrusion detection systems to protect data.	LANs only work within a small area like a building or office, so they are not suitable for long-distance connections.

Table 3: Benefits and constrains of Local Area Network(LAN)

Metropolitan Area Network (MAN)

A MAN is a type of computer network that covers a larger area than a LAN but smaller than a WAN. It usually connects devices across a city or a region. MAN uses technologies like ATM, Frame Relay, and MPLS, and high-speed connections such as fiber optic cables or leased lines. It is mainly used by businesses and organizations to connect different offices or locations within the same city or area.

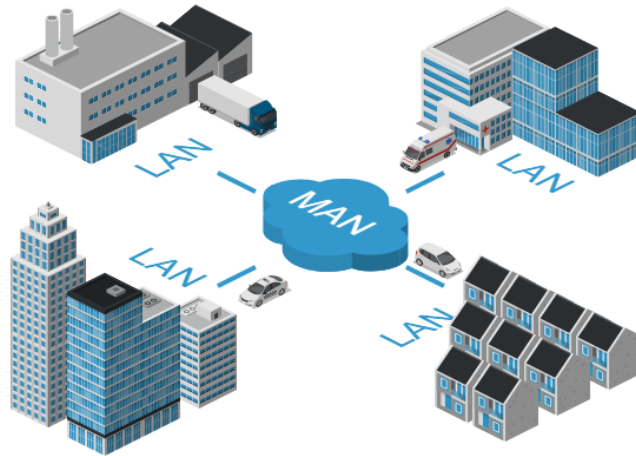


Figure 5: Metropolitan Area Network (MAN)

Benefits and constrains of MAN

Benefits	Constrains
MAN uses high-speed technologies like ATM, Frame Relay, and MPLS, so data transfer is fast and network performance is good.	MAN may not be suitable for very large organizations or those with rapidly growing users.
MAN can be easily expanded or reduced based on business needs, so it is suitable for growing organizations.	It can face security risks like hacking, malware, and phishing if not properly managed. Devices usually need physical connections, which can reduce mobility and flexibility.
It connects multiple locations within a city or region, helping businesses communicate and share resources easily.	Devices usually need physical connections, which can reduce mobility and flexibility.
MAN is more cost-effective than WAN when connecting locations within a city or region.	Setting up a MAN can be expensive, especially when using fiber optic technology.

Table 4: Benefits and constrains of Metropolitan Area Network (MAN)

Wide Area Network (WAN)

A WAN is a network that connects devices over a very large area, such as a city, country, or even different countries. It uses technologies like ATM, Frame Relay, and MPLS, and communication methods like leased lines and satellites. WAN is used to connect different LANs and helps businesses communicate across multiple locations.

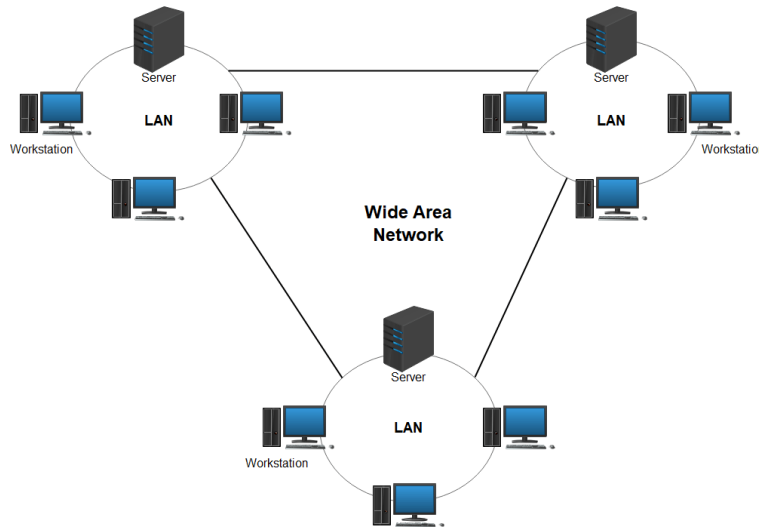


Figure 6: Wide Area Network (WAN)

Benefits and constrains of WAN

Benefits	Constrains
WAN connects devices over a very large area like cities or countries, helping organizations share resources and communicate easily.	WAN may have limited bandwidth, which can make the network slow.
WAN uses multiple communication paths, so the network can still work even if one path fails.	WAN is complex to set up and manage, especially over large areas.
WAN can be more cost-effective than using separate leased lines for each location.	WAN is vulnerable to security threats like hacking, malware, and phishing.

WAN is flexible and can be scaled up or down as a business grows.	WAN can be expensive to install, especially when using technologies like satellite or leased lines.
WAN can be secured using firewalls, encryption, and intrusion detection systems to protect data.	WAN may require physical connections, which can reduce mobility and flexibility.

Table 5: Benefits and constrains of Wide Area Network(WAN)

Campus Area Network (CAN)

A CAN is a network that connects multiple LANs within a single location, such as a university or industrial park. It allows devices in different buildings to communicate and share resources easily.

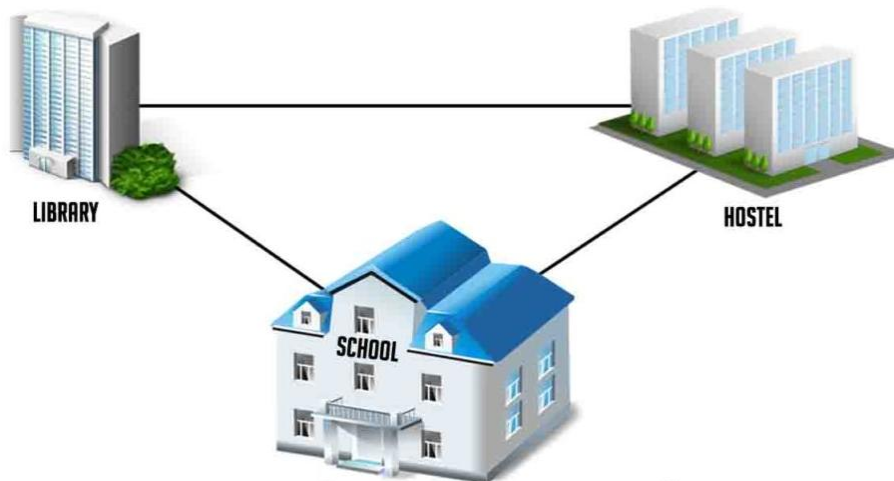


Figure 7: Campus Area Network (CAN)

Benefits and constrains of CAN

Benefits	Constrains
CAN allows communication and resource sharing between different LANs in the same location.	CAN can have signal problems like interference, which affects performance.
CAN allows users to access resources and services remotely.	CAN is complex and expensive to set up and maintain.
CAN provides high-speed communication between connected LANs.	CAN is difficult to update because technology changes quickly.

CAN is flexible and can be expanded when needed.	CAN needs careful planning, engineering, and high cost to build properly.
CAN offers centralized management and control of the network.	CAN is limited to a specific physical location.

Table 8: Benefits and constraints of Campus Area Network (CAN)

Storage Area Network (SAN)

A SAN is a high-speed network that connects storage devices, like disk arrays or tape libraries, to servers. It provides fast block-level access to data and usually uses protocols like Fiber Channel (FC) or iSCSI.

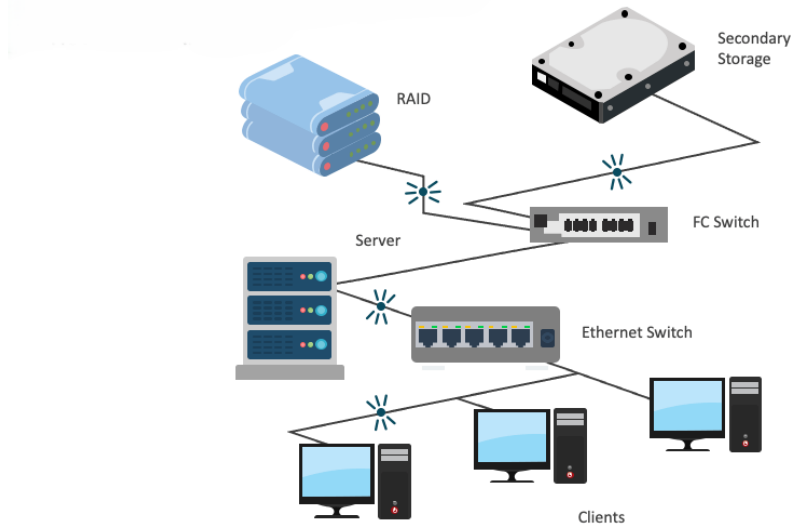


Figure 9: Storage Area Network (SAN)

Benefits and constraints of SAN

Benefits	Constraints
SAN provides advanced features like data replication and snapshots for better data availability and disaster recovery.	SAN is expensive to set up and operate due to specialized equipment and storage devices.
SAN can support large-scale storage and data-intensive applications, such as video streaming or business databases.	SAN is complex to implement, needing specialized hardware, software, and skilled professionals.

SAN has very high-speed data transfer for fast access and sharing.	SAN may have limited support from some operating systems compared to common network protocols.
SAN centralizes storage, making it easier to manage and protect data.	SAN can be difficult to integrate into an existing network.
SAN separates storage from servers, improving performance and scalability for both.	SAN is a potential security target, so strong security measures are essential.

Table 6: Benefits and constraints of Storage Area Network (SAN)

Virtual Private Network (VPN)

A VPN is a technology that creates a secure and encrypted connection over a public network, like the internet. It allows remote workers or users to safely access a company's network and resources, or protect personal online activity.



Figure 10: Virtual Private Network (VPN)

Benefits and constraints of VPN

Benefits	Constraints
Allows users to access blocked or restricted websites by routing traffic through servers in other countries.	May reduce internet speed due to encryption and server distance.
Provides enhanced security for online data and communication through encryption.	The VPN provider's jurisdiction can affect data privacy.

Protects user privacy by hiding the IP address.	Not all VPN services offer the same level of security and protection.
Can be a cost-effective solution for small and medium-sized businesses.	Requires additional setup and ongoing maintenance.
Enables remote access to a company's network and resources for remote workers	Limited number of server locations available

Table 7: Benefits and constraints of Virtual Private Network (VPN)

Network topology

Network topology is the layout or configuration of a computer network. It describes the connection of devices (such as computers, printers and servers) to each other and communication among devices in the network.

Physical topology

Physical topology is a description of the physical form of a network. It explains the physical connection of cables, devices, and other pieces of hardware.

Logical topology

Logical topology refers to how data flows within a network, regardless of the physical connections. It shows the path data takes from one device to another and how devices communicate logically.

Types of Physical topology

Bus topology

In bus topology, every device in the network is linked to one central and major cable known as a bus. This is a cable that serves as a common communication medium between all the devices. Bus is a straight (linear) cable that connects all the devices. It is also killed at the endings in order to prevent reflections of the signals and maintain the transmission of data. The connectors (not terminators) are used to connect each device to the main cable.

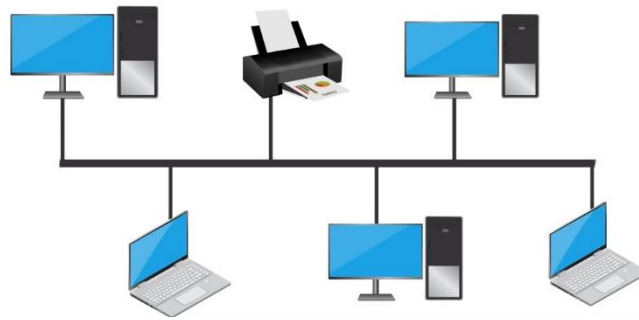


Figure 11: Bus topology

Ring topology

A ring topology is a topology where devices are linked in a circular (closed loop) form. Every device has two devices attached to it, one on both sides. The data flows in a single direction around the ring traversing every device until it gets to the destination.

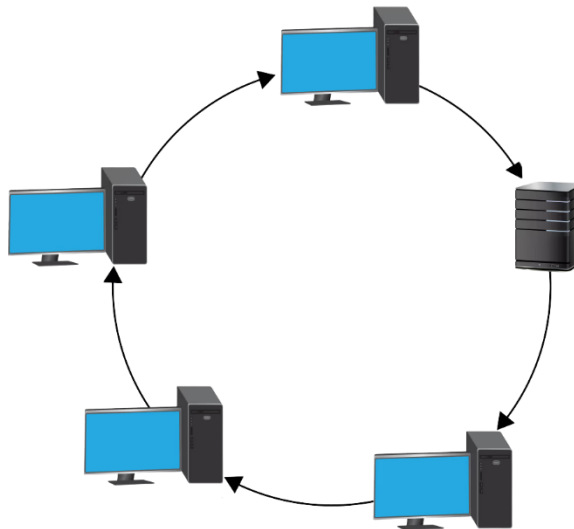


Figure 12: Ring topology

Star topology

A star topology has all the devices linked to a central switch or hub. This is the main device that governs and regulates the communication of all the devices that are connected. Information is transmitted through the hub then to the destination.

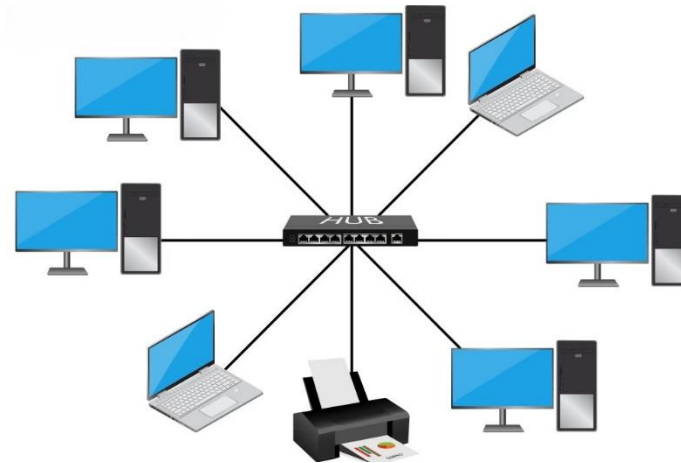


Figure 13: Star topology

Mesh topology

Mesh topology is a topology in which all the devices are linked to all the others. This provides various channels through which data can flow enhancing dependability. Every device is directly connected (point-to-point) with others.

Mesh topology has two types:

- Full Mesh: All the devices are connected with all other devices.
- Partial Mesh: Only some devices are connected to all others, not all.

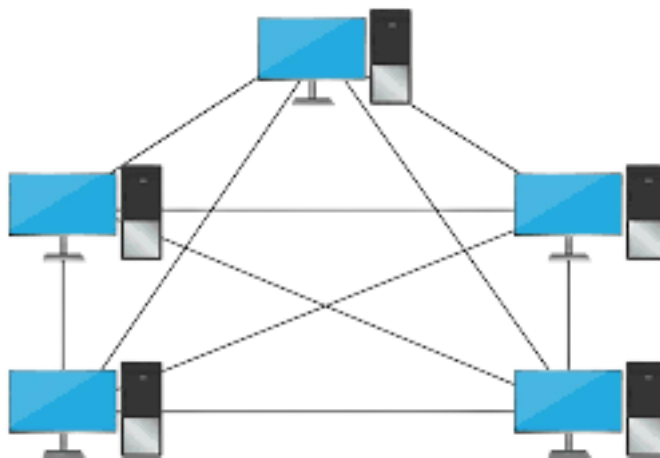


Figure 14: Mesh topology

Tree topology

A tree topology is a network topology which has a hierarchical (branch-like) design. It resembles an upside down tree where the main hub (root) is placed on top, and the associated branches, are placed in the lower part. The bottom devices are referred to as leaf nodes (end devices), whereas the intermediary ones are referred to as internal nodes or switches. The main node is that which links all the branches and determines the data circulation within the network.

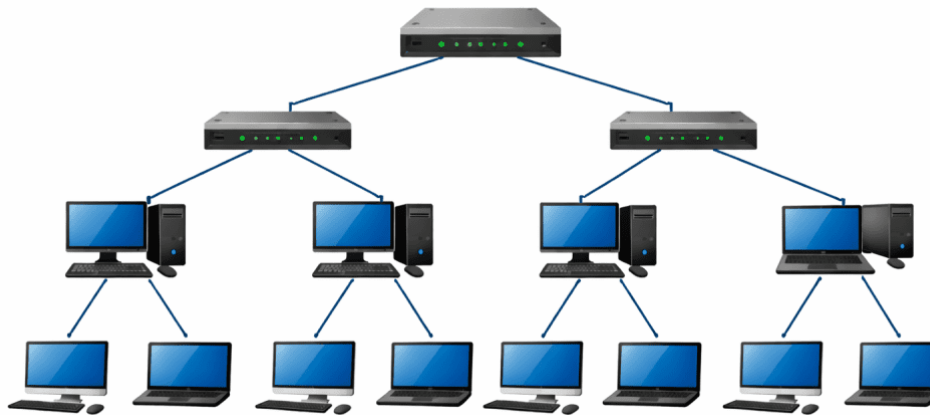


Figure 15: Tree topology

Hybrid topology

A hybrid topology consists of two or more network topologies. It is made to satisfy a particular requirement through the benefits of multiple topologies. To give an example, a star topology can be applied at the center of the network, and some other sections can have a ring topology. Hybrid topology uses the advantage of various topologies to offer flexibility, scalability and enhanced performance.

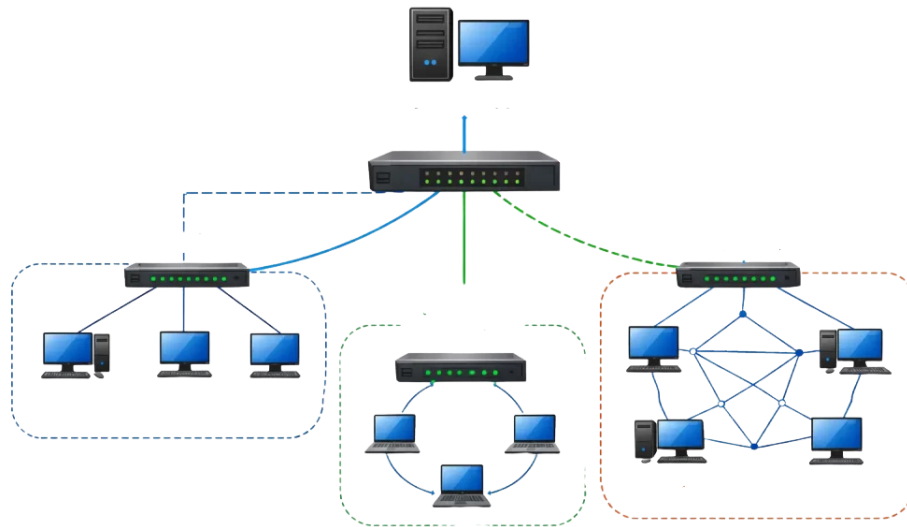


Figure 16: Hybrid topology

Benefits and Constrains of Physical topologies with comparison

Topology	Benefits	Constrains
Bus Topology	Simple to understand and implement	Limited by the total cable length, typically no more than several kilometers.
	Cost-effective, as only a single cable is needed to connect all devices.	Proper cable termination is required; otherwise, the network may fail.
	Does not require a central hub or switch.	Difficult to identify and troubleshoot issues, as all devices share the same cable
Ring Topology	Even if one device fails, the network can continue to function because data can be passed through the other devices	A ring topology is limited in its ability to expand, as adding new devices can cause disruptions to the network

	Data is not broadcasted to all devices, which reduces network traffic	It can be harder to troubleshoot and pinpoint the problem in a ring topology
	Each device on the ring can communicate directly with its two nearest neighbors	The bandwidth is shared by all devices on the ring, which can limit the overall performance of the network
Star Topology	Each device has its own direct (point-to-point) connection to the central hub, making it easy to add or remove devices.	Requires more cables and a central hub, which can increase cost.
	Failure of one device or connection does not affect the rest of the network.	If many devices send data at the same time, the hub can become overloaded, causing slow performance.
	The central hub can control access, providing better network security.	The number of devices is limited by the capacity of the hub.
Mesh Topology	Each device has a dedicated point-to-point link to every other device, resulting in increased fault tolerance and the ability to self-heal the network in case of a link failure.	The number of connections required grows rapidly as the number of devices in the network increases, leading to increased complexity and management overhead.
	The decentralized nature of mesh topology means that	The cost of implementing a mesh network can be high,

	there is no single point of failure, making the network highly resilient	as each device requires multiple connections to other devices.
	Each device can communicate directly with every other device, reducing the need for network traffic to pass through intermediate devices.	Mesh networks can be more difficult to set up and configure than other network topologies, such as star or bus.
Tree Topology	Provides a hierarchical structure, making large networks easy to organize and manage.	The central hub can become a bottleneck if many branches communicate at the same time.
	Can be easily expanded by adding more branches (scalable and modular).	Requires additional cabling to connect the hub and branches.
	Makes it easier to identify faults due to its structured design.	Can become expensive as more branches and connections are added.
Hybrid Topology	Combines advantages of different topologies.	More complex to design and manage.
	Highly flexible and scalable for different network needs.	Requires extra configuration and maintenance.
	Improves performance by using suitable topology in each part.	Can be expensive to implement.

Table 8: Benefits and Constrains of Physical topologies with comparison

Types of logical topology

Ethernet

Ethernet is a widely used networking standard that commonly uses a bus logical topology. In Ethernet networks, data is transmitted over a shared medium such as coaxial cable or twisted pair cables.

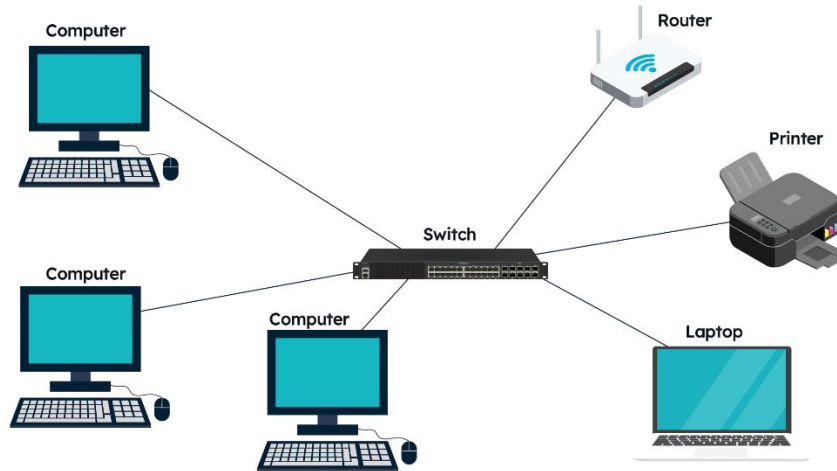


Figure 17: Ethernet

Token Ring

Token Ring is a logical topology that uses a ring structure to control data transmission. In this network, a special packet called a “token” circulates around the devices in the ring. Each device checks the token when it receives it. If the device has data to send, it attaches the data to the token and passes it to the next device in the ring. This method ensures orderly and efficient communication in the network.

Local Talk

Local Talk is a proprietary networking protocol developed by Apple. It uses a bus logical topology and was commonly used in Macintosh computers during the 1980s and 1990s. Local Talk uses twisted pair cables for communication. It also uses a method similar to Carrier Sense Multiple Access with Collision Detection (CSMA/CD), which allows multiple devices to share the same cable and manage data transmission efficiently.

Asynchronous Transfer Mode (ATM)

Asynchronous Transfer Mode (ATM) is a type of logical topology that is used for high-speed data transmission. ATM is a cell-based switching technology, which means it divides data into small fixed-size cells before transmission.

Impact of Network Topology in Computer Networks

Network topology has a major impact on the performance, security, and scalability of a network.

- **Reliability:** Redundant topologies like ring and mesh provide multiple paths for data. This helps the network continue working even if a device or connection fails.
- **Performance:** A topology that reduces distance and number of hops between devices can lower delay (latency) and improve overall network speed.
- **Scalability:** Hierarchical topologies like star and tree make it easier to add or remove devices without affecting the whole network.
- **Security:** Topologies that separate or isolate network sections (like VLANs) improve security by reducing the number of possible attack points.

Networking Standards

Networking standards are a set of technical rules and guidelines used to design, build, and maintain computer networks. They define the protocols, technologies, and practices that ensure devices in a network can communicate and share resources properly. These standards help different devices and systems work together (interoperability), allowing smooth data exchange and supporting the growth and development of networks over time.

International Networking Standards Organizations

There are several international organizations responsible for developing and maintaining networking standards. These standards help ensure devices and networks work together properly around the world.

The Internet Engineering Task Force (IETF) - The IETF is an international organization that develops and publishes standards for the internet. It is responsible for the development of a wide range of internet-related protocols, including the Transmission Control Protocol (TCP) and the Internet Protocol (IP), which form the foundation of the internet.

The International Organization for Standardization (ISO) - ISO is a global organization that develops and publishes standards across a wide range of industries, including information technology. ISO's work includes the development of networking standards such as the Open Systems Interconnection (OSI) model, which provides a framework for understanding the different layers of a network.

The Institute of Electrical and Electronics Engineers (IEEE) - IEEE is a professional organization that develops and publishes standards for a wide range of technologies, including networking. Its 802 standards for local area networks (LANs), wide area networks (WANs), and wireless networks, such as Ethernet and Wi-Fi, are widely used in networking industry.

The Telecommunications Industry Association (TIA) - TIA is a trade association that develops and publishes standards for the telecommunications industry. Its standards cover a wide range of areas, including wireless communications, networking, and broadband technologies.

IEEE standards in computer networking

IEEE standards in computer networks	Description
IEEE 802	It is used for the overview and architecture of LAN/MAN.
IEEE 802.1	It is used for bridging and management of LAN/MAN.
IEEE 802.1 s	It is used in multiple spanning trees.
IEEE 802.1 w	It is used for rapid reconfiguration of spanning trees.
IEEE 802.1 x	It is used for network access control of ports.
IEEE 802.2	It is used in Logical Link Control (LLC).
IEEE 802.3	It is used in Ethernet (CSMA/CD access method).
IEEE 802.3ae	It is used for 10 Gigabit Ethernet.

IEEE 802.4	It is used for token passing bus access methods and the physical layer specifications
IEEE 802.5	It is used for token ring access methods and the physical layer specifications.
IEEE 802.6	It is used in distributed Queue Dual Bus (DQDB) access method and for the physical layer specifications (MAN).
IEEE 802.7	It is used in broadband LAN
IEEE 802.8	It is used in fiber optics.
IEEE 802.9	It is used in isochronous LANs.
IEEE 802.10	It is used in interoperable LAN/MAN security.
IEEE 802.11	It is used in wireless LAN, MAC, and Physical layer specifications.
IEEE 802.12	It is used in the demand-priority access method, in the physical layer, and in repeater specifications.
IEEE 802.13	It is not used.
IEEE 802.14	It is used in cable modems (not used now).
IEEE 802.15	It is used in WPAN (Wireless Personal Area Network).
IEEE 802.16	It is used in Wireless MAN (Wireless Metropolitan Area Network).
IEEE 802.17	It is used in RPR access (Resilient Packet Ring).

Table 9: IEEE standards in computer networking

Data Communication

Data communication in a network refers to the process of sending and receiving data between devices and systems that are connected to a network. It is the basic foundation of all network systems and allows users to share information, resources, and services.

Communication in a network can happen in different ways:

- Peer to peer -This occurs when two or more devices communicate directly with each other without using a central server or intermediary.

- Client-server -This occurs when a client device requests data or services from a server, and the server responds to the request. Broadcast
- Broadcast -This occurs when a message is sent to all devices in the network instead of a specific device or group.
- Communication Across the OSI Model - Communication in a network happens across different layers of the OSI model, and each layer has specific protocols and functions.
- Application Layer -Protocols like HTTP, FTP, and SMTP manage communication between applications and the network.
- Transport Layer-Protocols like TCP and UDP manage communication between devices and ensure data is delivered accurately and reliably.
- Network Layer -Protocols like IP (Internet Protocol) handle communication between different networks by routing data to the correct destination.

Transmission Modes

Transmission modes refer to the ways in which data is transmitted from a source to a destination in a communication system. Some common types of transmission modes include,

- Simplex mode - Data is transmitted in one direction only.
- Half-duplex mode - Data can be transmitted in both directions, but not at the same time.
- Full-duplex mode - Data can be transmitted in both directions simultaneously.
- Multiplexing - Multiple signals are combined into a single signal for transmission.
- Broadcasting - Data is transmitted to all devices on a network at the same time.
- Point-to-point mode - Data is transmitted between two devices without passing through any intermediate devices.
- Point-to-multipoint mode - Data is transmitted from a single device to multiple devices simultaneously

The transmission mode used depends on the requirements of the network and the type of data being transmitted. In general, full-duplex mode is commonly used in high-speed networks because it allows data to be sent and received at the same time. Simplex and half-duplex modes are usually used in lower-speed networks where simultaneous communication is not required

Guided media and Unguided media

Guided and unguided media are two types of communication channels used to transmit data from one place to another. The main difference is whether a physical medium is used to carry the signal.

Guided media

Guided media refers to communication channels that use a physical pathway to transmit signals. Examples include twisted pair cables, coaxial cables, and fiber optic cables. These media provide a direct and fixed path for data transmission, which makes the connection more stable and reliable.

Examples of Guided Media

- **Twisted Pair Cables** - Twisted pair cables are copper wires commonly used in Ethernet networks. They consist of two copper wires twisted together to reduce signal interference.
- **Coaxial cables** - Coaxial cables are widely used in cable TV and broadband internet connections. They have a copper core surrounded by insulation and a metal shielding layer for protection.
- **Fiber optic cables** - Fiber optic cables are thin, flexible cables that transmit data using light signals. They are made of glass or plastic fibers and are used for high-speed and long-distance communication.

Unguided Media

Unguided media refers to communication channels that do not use a physical medium (like cables) to transmit signals. Instead, data is sent through the air or space using signals such as radio waves, infrared, and satellite communication. These signals travel freely through the environment and may be affected by interference, obstacles, or weather conditions.

- **Radio Waves** - These are electromagnetic waves that are used for wireless communication, such as Wi-Fi and mobile networks.

- Infrared - This type of light is used for short-range communication, such as remote controls for televisions and other electronic devices.
- Satellite Communication - This uses satellites in space to transmit signals to and from the Earth. It is commonly used for communication in remote areas, or for global communication.

Bandwidth

Bandwidth is the amount of data that can be sent through a communication channel within a specific time. It is usually measured in bits per second (bps) or bytes per second. A higher bandwidth means more data can be transferred faster over a network.

Bandwidth Requirements

Bandwidth requirements refer to how much data capacity is needed for an application, service, or system to work efficiently. These requirements depend on the type of application and the number of users using it at the same time.

Procedures to Increase Bandwidth

There are several methods to improve bandwidth and overall network performance,

- Optimize network topology - By optimizing the layout of the network, it is possible to reduce congestion and increase bandwidth.
- Use Quality of Service (QoS) - Prioritize important data (like video calls) so it gets faster and more reliable transmission.
- Offload traffic - Shift some network traffic to external services such as cloud systems to reduce local network load.
- Increase the number of channels - Use multiple communication paths to distribute data and improve speed.
- Upgrade cables - Replace old cables with high-performance ones like Cat6 or Cat7 to support higher bandwidth.
- Add more access points - Improve wireless coverage and allow more devices to connect efficiently.

- Use network compression - Reduce the size of data before sending it to save bandwidth.
- Implement load balancing - Distribute traffic across multiple servers or devices to avoid overload and increase total capacity.

Bandwidth vs Speed

In Computer Networks capacity is defined as bandwidth and transfer rate is defined as speed. More bandwidth doesn't mean better performance.

Bandwidth	Speed
Bandwidth is the maximum amount of data that can be transmitted over a network	Speed is the actual rate at which data is transferred.
Measured in bits per second (bps).	Also measured in bits per second (bps).
Represents the theoretical capacity of the network.	Represents real-time performance.
Determined by network infrastructure	Affected by congestion, number of users, and device performance.
Usually constant.	Can vary depending on conditions.
Refers to total capacity.	Refers to usage of that capacity.

Table 10: Bandwidth vs Speed Comparison

Network Models

Network models are frameworks used to understand and organize how computer networks work. They break down the complex process of data communication into smaller, manageable layers and explain how each part interacts with others. Some of the most commonly used network models are the OSI Model and the TCP/IP Model. These models are not limited to a specific type of network but can be applied to various networks such as LANs, WANs, and the Internet.

OSI Model

The OSI Model is a theoretical framework used to explain how data is transmitted over a network. It acts as a reference model that defines the functions of a networking system and describes how different devices communicate using standard protocols.

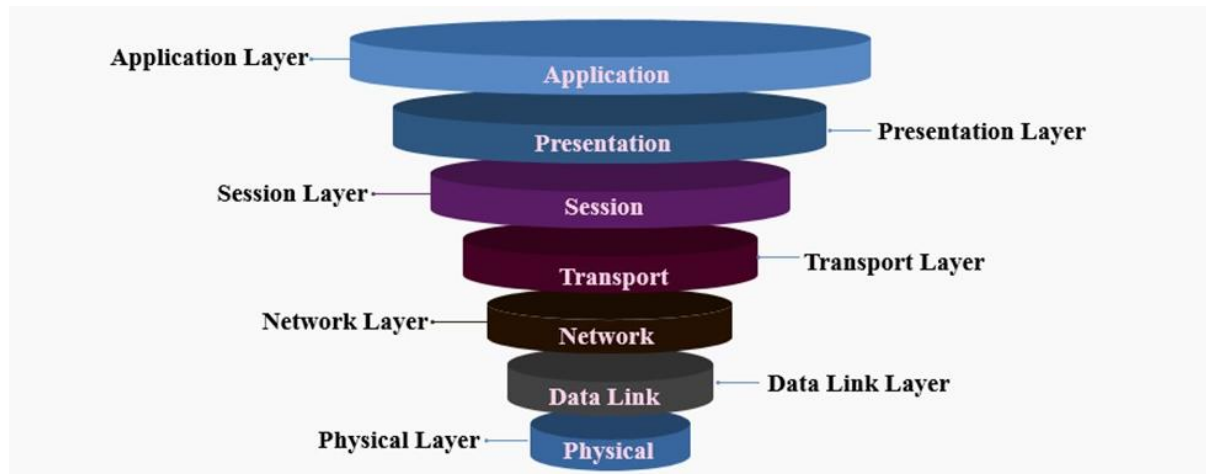


Figure 18: OSI Model

Layers of the OSI Model

- Physical Layer - Handles the physical connection between devices, including cables, connectors, and signal transmission.
- Data Link Layer - Ensures a reliable connection between two devices. It manages error detection, correction, and flow control.
- Network Layer - Responsible for routing data across networks. It determines the best path and handles device addressing.
- Transport Layer - Ensures data is delivered correctly and in order. It segments data and reassembles it at the destination.
- Session Layer - Establishes, maintains, and terminates communication sessions between devices. It also synchronizes data exchange.
- Presentation Layer - Translates data into a format understandable by the application. It also handles encryption and compression.

- Application Layer - Acts as the interface between the user and the network, providing access to network services.

TCP/IP Model

The TCP/IP Model is a set of communication protocols used to transmit data across the internet and other networks. It forms the foundation of the modern internet. Compared to the OSI Model, it is simpler and more practical for real-world use.

Layers of the TCP/IP Model

- Link Layer - Handles the physical connection between devices, including cables and hardware. It combines the functions of the Physical and Data Link layers of the OSI Model.
- Internet Layer - Responsible for routing data between devices across networks. It finds the best path and manages addressing. This corresponds to the Network Layer of the OSI model.
- Transport Layer - Ensures reliable data delivery in the correct order. It breaks data into segments and reassembles them at the destination. This matches the Transport Layer of the OSI model.
- Application Layer - Acts as the interface for users to access network services. It combines the functions of the Session, Presentation, and Application layers of the OSI model.

Comparison between OSI and TCP/IP Model

OSI Model	IP Model
The OSI model is a reference model that describes the functions of a networking system and the protocols used to connect different devices	The TCP/IP model is a set of communication protocols used to transmit data over the internet and other networks.

It consists of seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application.	It consists of four layers: Link Layer, Internet Layer, Transport Layer, and the Application Layer.
Each layer communicates with the layer directly above and below it, and provides services to the layer above it and receives services from the layer below it.	The most fundamental protocols of the TCP/IP model are the Internet Protocol (IP) and the Transmission Control Protocol (TCP).
Data is passed from one layer to the next as it travels through the network, with each layer adding its own header to the data packet.	The TCP/IP model is extensible, allowing for additional protocols and technologies to be added as needed.
The OSI model is not a protocol itself, rather it's a way of describing the various functions and protocols that are required to build an effective network.	TCP/IP has become the standard protocol suite of the internet, many LANs and WANs, and many embedded systems.

Table 11: Comparison between OSI and TCP/IP Model

Protocols

A network protocol is a set of rules and standards that control how data is sent and received over a network. It defines how data packets are structured and how communication takes place between devices.

Transmission Control Protocol (TCP)

The Transmission Control Protocol is one of the main protocols in the Internet Protocol (IP) suite. It is a reliable, connection-oriented protocol that creates a virtual connection between two devices to ensure data is delivered correctly.

Transmission Control Protocol guarantees accurate data delivery by using flow control and error checking. It controls the amount of data sent using a window-based system and retransmits data if any errors are detected until it is received correctly.

Internet Protocol (IP)

The Internet Protocol is the main protocol used to send data across the internet and other networks in the TCP/IP Model. It is responsible for routing data packets between devices such as computers and servers.

There are two main versions of IP:

- IPv4 - A 32-bit addressing system written in dotted-decimal format.
- IPv6 - A 128-bit addressing system written in hexadecimal format separated by colons.

User Datagram Protocol (UDP)

The User Datagram Protocol is a simple, connectionless protocol that allows data to be sent over an IP network without establishing a connection first. Unlike TCP, it does not guarantee delivery, order, or error checking, but it provides faster communication.

Post Office Protocol (POP)

The Post Office Protocol (POP) is used to retrieve email messages from a mail server to a client device. It downloads emails from the server and usually deletes them after saving a copy on the client. It is one of the main email retrieval protocols, along with IMAP.

Simple Mail Transfer Protocol (SMTP)

The Simple Mail Transfer Protocol is used for sending email messages between servers. It transfers emails from one server to another and delivers them to email clients like Outlook or Apple Mail. It is a text-based protocol and is the standard method for sending emails over the internet. SMTP is mainly used for sending mail, while IMAP is used for receiving and managing emails on the server.

File Transfer Protocol (FTP)

The File Transfer Protocol is a standard network protocol used to transfer files between a client and a server over a network such as the internet. It can also be used within the same system to move files between devices or locations.

Hypertext Transfer Protocol (HTTP)

The Hypertext Transfer Protocol is used to transfer data over the internet and forms the foundation of the World Wide Web. It works on a request-response model, where a browser sends a request and the server returns the required data.

HTTP is not encrypted, so data can be seen by others. To improve security, HTTPS was created, which uses encryption technologies like TLS or SSL to protect data during transmission.

Telnet

Telnet is a protocol that allows remote access to another computer using a command-line interface. It lets a user control a remote system as if they were physically using it.

Secure Sockets Layer (SSL)

The Secure Sockets Layer (SSL) and its modern version Transport Layer Security (TLS) are security protocols used to encrypt communication over the internet. They create a secure connection between a web browser and a web server to protect data from being intercepted.

Address Resolution Protocol (ARP)

The Address Resolution Protocol is used to map an IP address to a physical (MAC) address on a local network. It helps identify the hardware address of a device so that data can be delivered to the correct device within the network.

Dynamic Host Configuration Protocol (DHCP)

The Dynamic Host Configuration Protocol is used to automatically assign IP addresses and other network settings to devices on a network. It provides information such as subnet mask, default gateway, and DNS servers, making network setup easier and more efficient.

Internet Message Access Protocol (IMAP4)

The Internet Message Access Protocol is used to access and manage email messages directly on a mail server. Unlike POP, it does not download and remove emails from the server, allowing users to view and organize messages from multiple devices.

Session Initiation Protocol (SIP)

The Session Initiation Protocol is used to start, manage, and end real-time communication sessions such as voice calls, video calls, and messaging over IP networks. It supports both one-to-one and group communication services.

Protocol's Role in Effectiveness in a Network System

Protocols are really important for the network system of Redco Developments, a company that makes applications and has its main office in Kandy and a new branch in Galle. Protocols are like a set of rules that help devices, departments and branches talk to each other even if they use different equipment, operating systems or network technologies.

For the network design of Redco Developments we need protocols like TCP/IP, DHCP, DNS, HTTP/HTTPS, FTP and VPN to make sure everything runs smoothly between the Kandy and Galle offices. These protocols help departments like Developers, HR, Finance, Sales & Marketing and IT Support share information use applications and communicate with each other over the network.

Protocols also make sure that information is sent reliably and accurately. For example TCP makes sure that pieces of information are delivered in the order and are not lost, which is really

important for teams working on big mobile app projects. On the hand UDP is good for things like video meetings in the boardroom where we need fast communication.

Protocols like DHCP also make network management easier by giving IP addresses to all devices in both the Kandy and Galle branches, which reduces mistakes and the amount of work administrators have to do. DNS helps users find external services easily by turning domain names into IP addresses.

Security is also a role of protocols in this case. HTTPS and VPN protocols keep communication between the two branches by encrypting information and protecting it from people who should not have access to it. This is really important for Redco Developments because the company handles information about client applications and business.

Moreover protocols support something called Quality of Service which makes sure that different types of network traffic are treated differently based on how important they're. For example video meetings, in the boardroom and Galle administration meetings need to be treated as high priority while sending files between departments can use standard internet speed.

In general protocols are the foundation of the Redco Developments network system making sure that all departments and branches can work together smoothly reliably securely and efficiently. Redco Developments relies on protocols to keep its network system running well. Protocols are essential for Redco Developments to function properly.

Conclusion on Topology, Protocols, and Network Types Selected

For Redco Developments we have chosen a hybrid network design to meet the needs of the head office and the new Galle branch. This design uses Star topology within each department and floor. It also uses a WAN connection with VPN between the Kandy and Galle offices.

The Star topology is really good for both offices because it connects all devices like computers, printers, IP cameras and smart systems to a switch. This makes it easy to manage, troubleshoot

and expand departments like Developers, HR, Finance and Administration. If a device fails only that device is affected so Redco Developments can keep working with disruption.

We also think an Ethernet-based network design is an idea for internal communication within both branches. Ethernet is fast, reliable and works with the devices Redco Developments uses, such as smart devices, ERP systems and IP camera systems. It also supports growth, which's important for Redco Developments as it expands in the Southern province.

For communication between the Kandy and Galle branches we recommend using a WAN connection with VPN. This lets the branches communicate securely and privately over the internet. It means departments in both branches can access shared resources like servers, databases and applications safely and efficiently.

Redco Developments also needs LAN, WLAN and VLAN technologies. LAN provides secure communication within departments WLAN gives users like the Sales and Marketing team and the Customer Service team the freedom to move around and VLAN keeps the network secure by dividing it into separate parts as needed.

In the end using Star topology WAN, with VPN and an Ethernet-based LAN structure gives Redco Developments a network that's scalable, secure and high-performance. This design supports communication integrates smart devices and allows for future growth while keeping Redco Developments running smoothly and securely across both the Kandy and Galle branches.